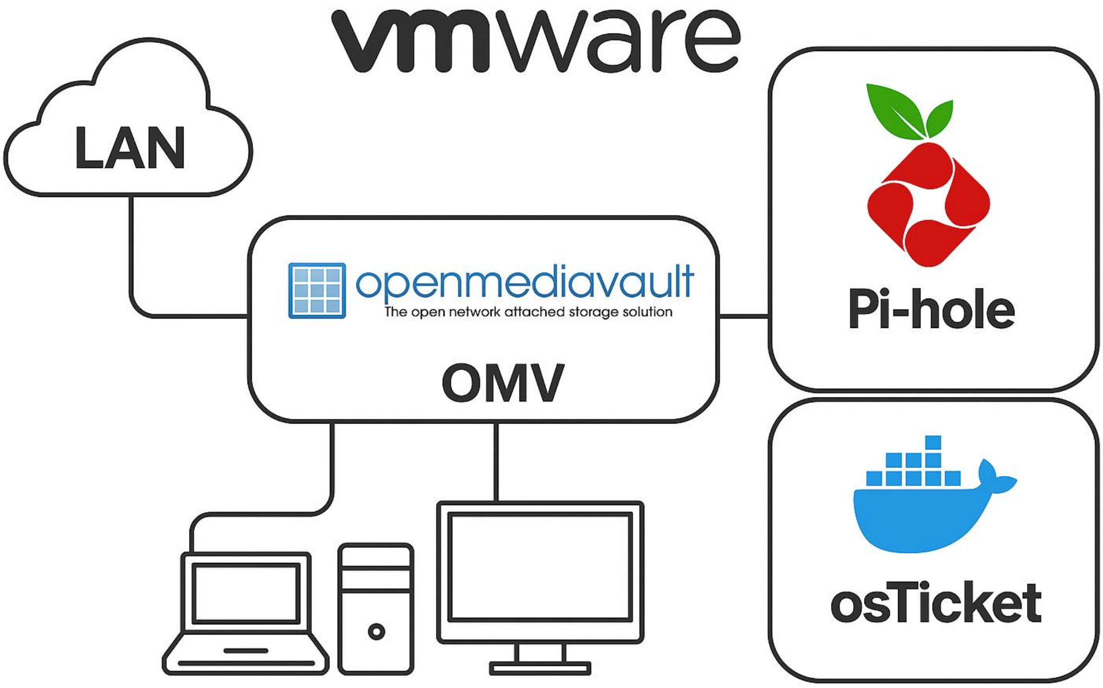
Introduction to infrastructures

Assignments Mac Appendices 2025 - 2026

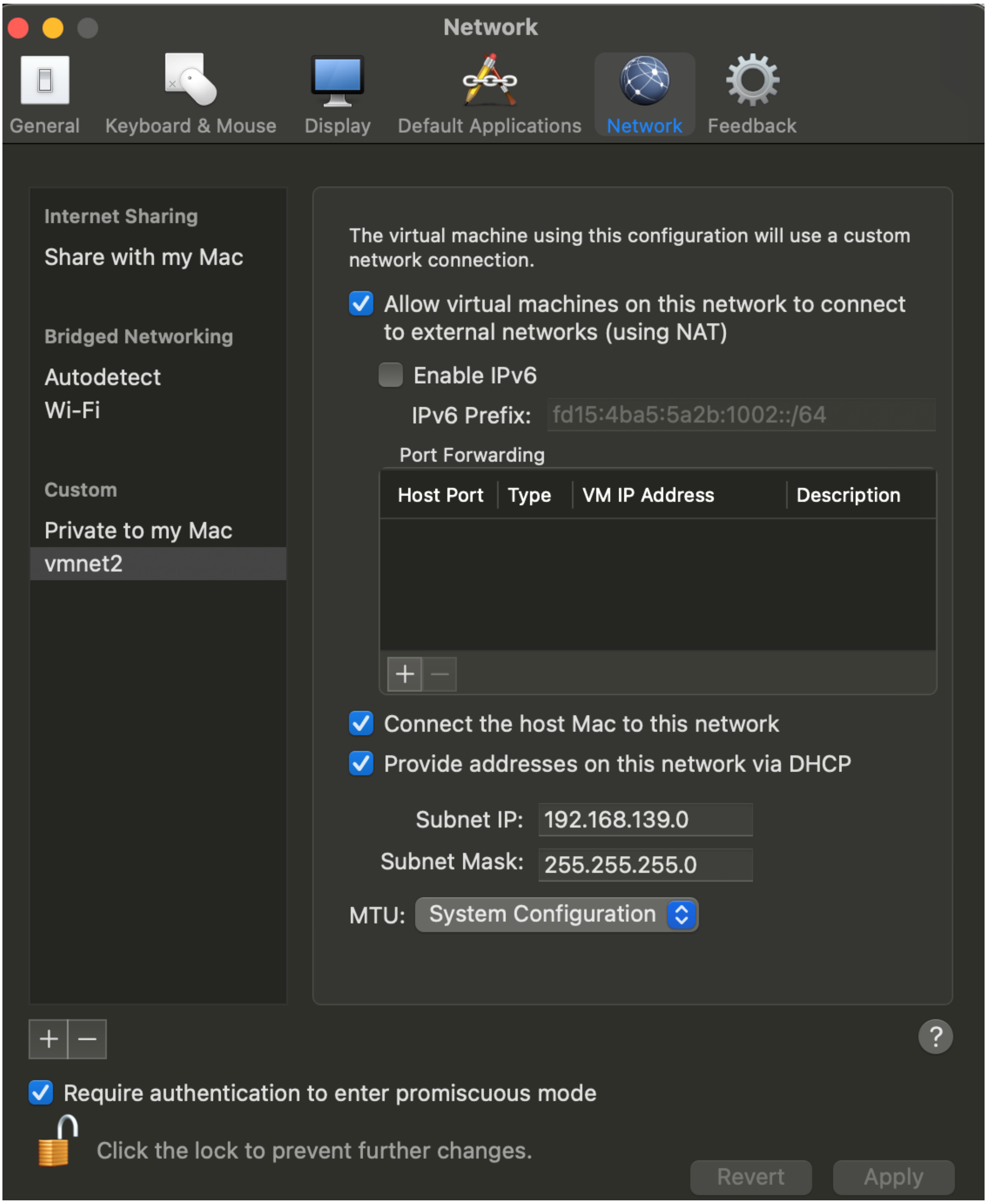


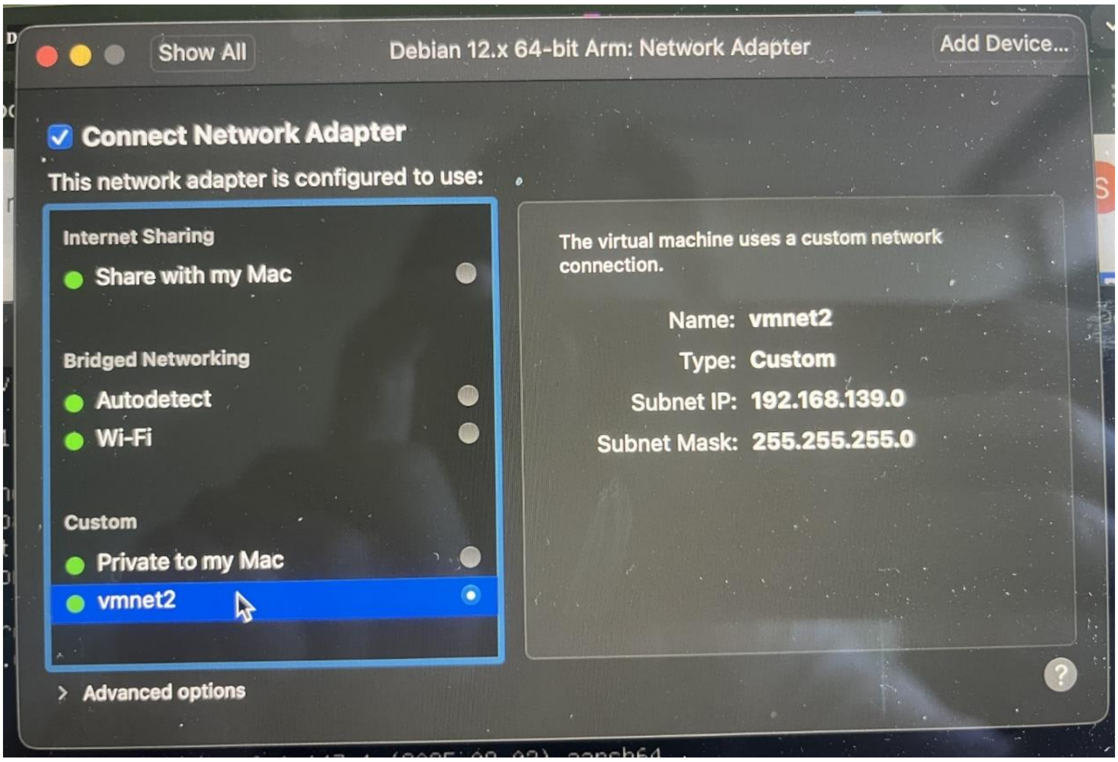
Version: 20250914.0311.MSI11

Contents

Week 1 - Mac

1.1 VMware Fusion Pro network settings





Screenshots provided by: Sofiia Ivanova

Week 2 - Mac

No differences

Week 3 - Mac

3.3 Installing osTicket app via docker compose

osTicket docker compose yaml file (**Testing and debugging done by:** David Denis):


```yaml
---
services:
 osticket:
  image: tiredofit/osticket:3.5.2
  container_name: OsTicket
  depends_on:
   - osticket_db
  ports:
   - "8080:80"  # Maps host port 8080 to container port 80
  networks:
   backend-network:
  environment:
   # Database configuration
   DB_HOST: osticket_db
   DB_PORT: 3306
   DB_NAME: my_database_name
   DB_USER: some_user
   DB_PASS: some_password
   # Admin configuration
   ADMIN_USER: mike
   ADMIN_PASS: Welkom#01
   ADMIN_EMAIL: yourown@email
   ADMIN_FIRSTNAME: Mike
   ADMIN_LASTNAME: Sierra
   # Installation config
   INSTALL_SECRET: mysecret
   INSTALL_URL: http://osticket.local
   INSTALL_NAME: Helpdesk
   CRON_INTERVAL: 1
  volumes:
   - CHANGE_TO_COMPOSE_DATA_PATH/osticket/config:/var/www/html
  restart: unless-stopped

 osticket_db:
  image: mariadb:11.3-jammy
  container_name: OsTicket_DB
  security_opt:
   - no-new-privileges:true
  environment:
   MYSQL_ROOT_PASSWORD: root_password
   MYSQL_USER: some_user
   MYSQL_PASSWORD: some_password
   MYSQL_DATABASE: my_database_name
  volumes:
   - CHANGE_TO_COMPOSE_DATA_PATH/osticket/db:/var/lib/mysql:rw
  networks:
   - backend-network
  restart: unless-stopped

networks:
 backend-network:
  name: osticket-backend-network
```


See week 3 in the intro infra app for the docker compose yaml file. The IP of Pi-hole is now: 192.168.139.11:8080. Same IP as the OpenMediaVault server but on another port.

Week 4 – Mac

On a Mac we are not going to use the docker container for Pi-hole. We need to create a new virtual machine and install Pi-hole on that.

4.1 Debian 12 Server in VMware for Pi-hole

- Install a new Debian 12 Server VM
- 1cpu, 2cores, 2GB ram, 32GB hard disk, NAT network adapter, CD/DVD drive.
- ISO: AMD64, ARM64
- See installation video in the intro infra app at week 1.
- Use hostname: pihole<your student number>
- Domain: local
- Don’t create a root user, create a user with sudo privileges that has your first name.
- Don’t install a Graphical desktop environment

Select SSH server and standard system utilities only!

Finish the Debian 12 installation and let it reboot.

Install Pi-hole on Debian 12 Pi-hole VM

Step 1: Update your system

Use the following terminal commands:

```
sudo apt update && sudo apt upgrade -y
sudo apt install curl sudo -y
```

Step 2: Identify your network interface

Use the following terminal command:

```
ip a
```

Write down the actual network interface name. For example: ens33.

Step 3: Configure a static IP using systemd-networkd

Create a new network config file for your interface:

```
sudo nano /etc/systemd/network/20-static.network
```

Example for interface eth0, paste the following in the configuration file:

```
[Match]
# Change this to match your network interface name (ens18, enp0s3, etc.)
Name=eth0

[Network]
Address=192.168.139.3/24
Gateway=192.168.139.2
DNS=192.168.139.2
```

Enable and restart systemd-networkd:

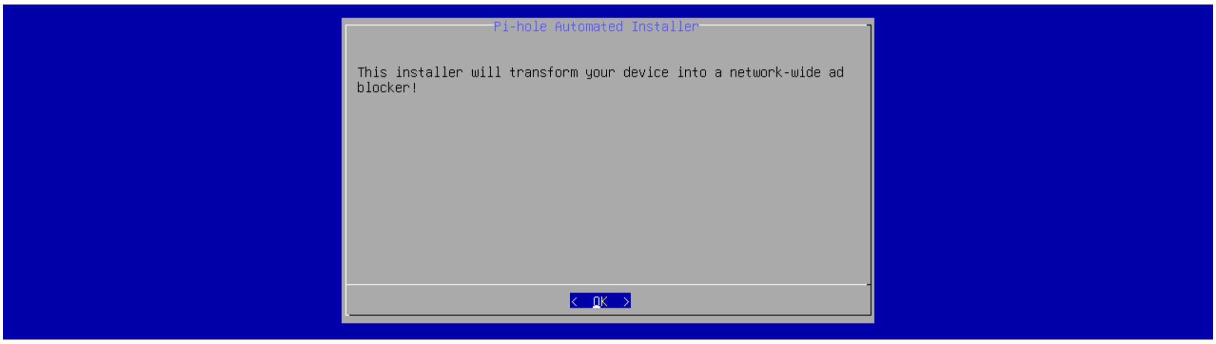
Use the following terminal commands:

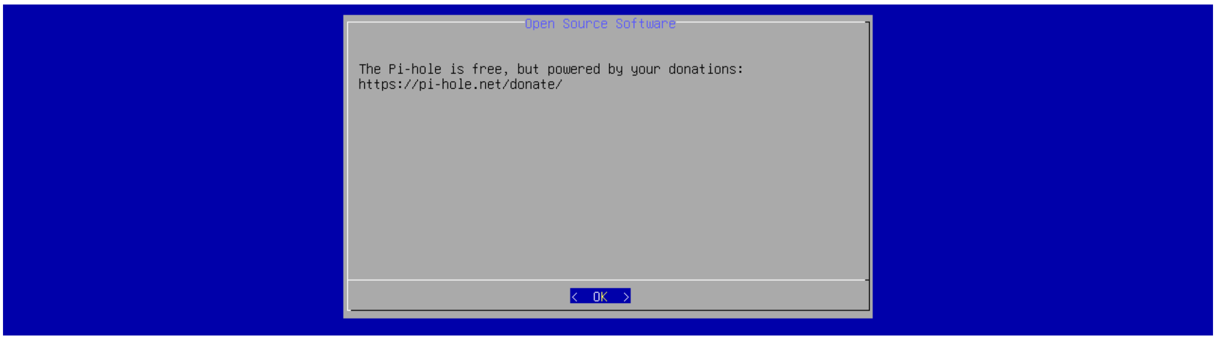
```
sudo systemctl enable systemd-networkd
sudo systemctl restart systemd-networkd
```

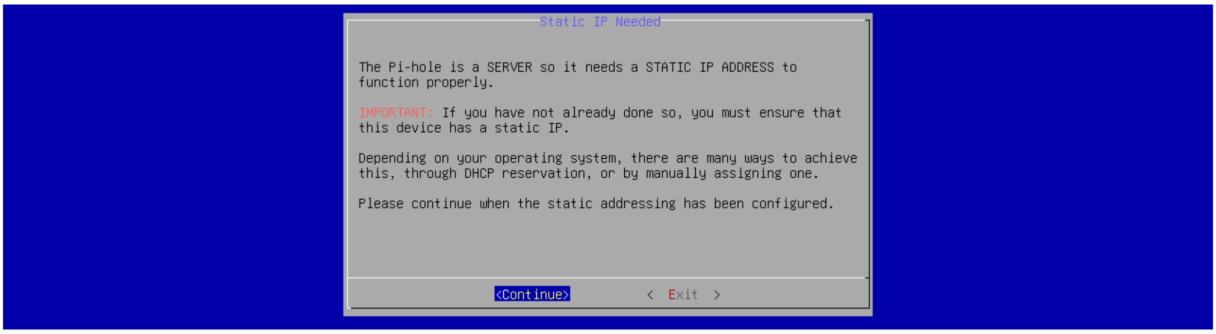
Step 4: Install Pi-hole

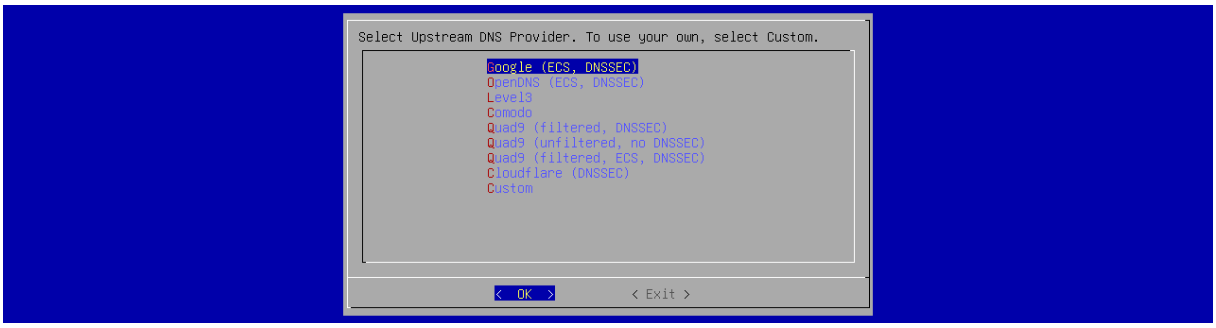
Use the following terminal command:

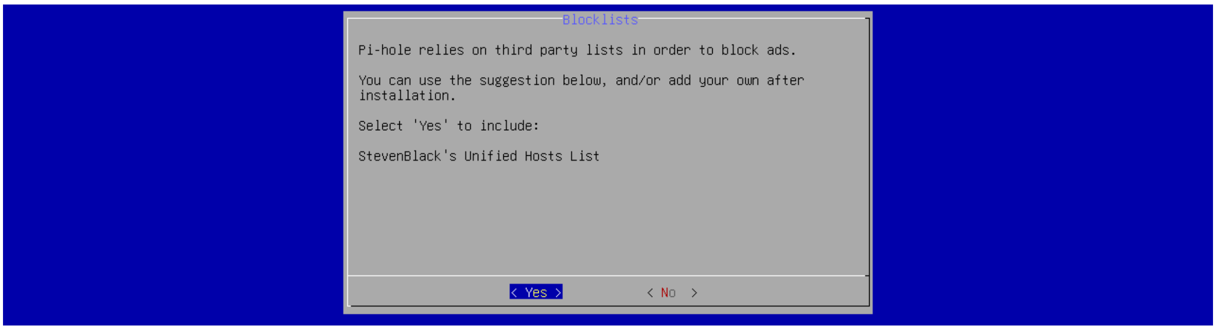
```
curl -sSL https://install.pi-hole.net | sudo bash
```

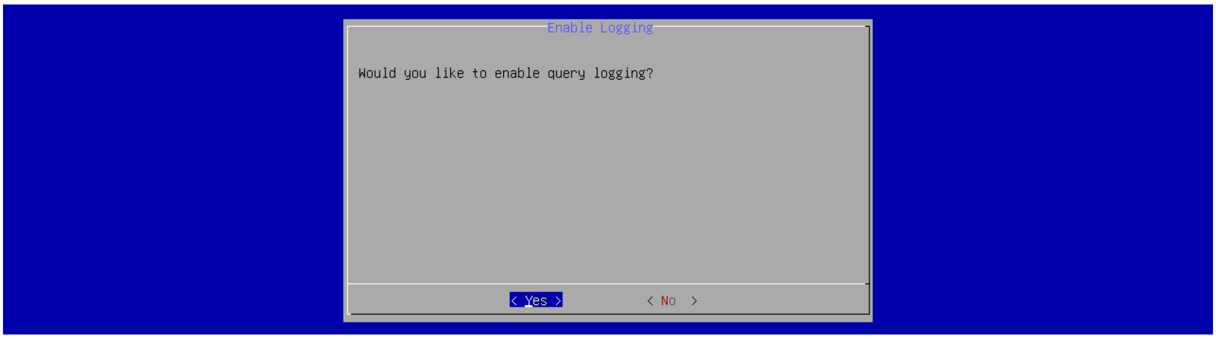


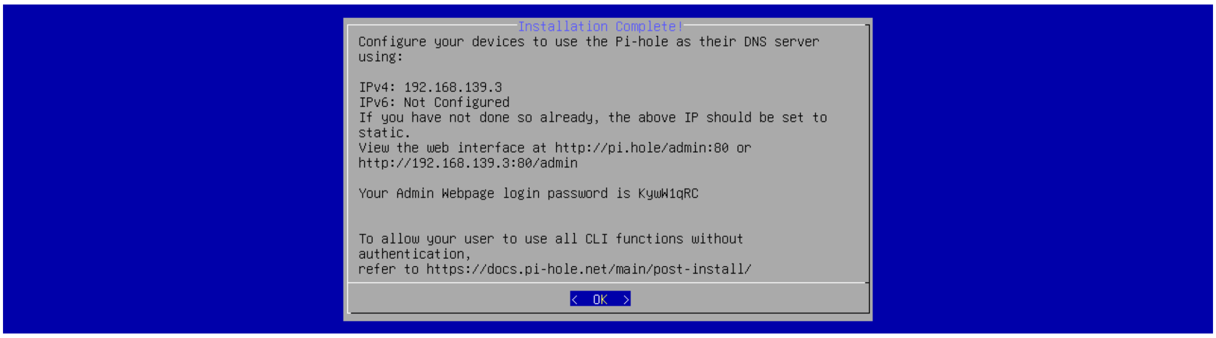

Open Source Software
The Pi-hole is free, but powered by your donations:
https://pi-hole.net/donate/
< OK >


Static IP Needed
The Pi-hole is a SERVER so it needs a STATIC IP ADDRESS to function properly.
IMPORTANT: If you have not already done so, you must ensure that this device has a static IP.
Depending on your operating system, there are many ways to achieve this, through DHCP reservation, or by manually assigning one.
Please continue when the static addressing has been configured.
<Continue>
< Exit >


Select Upstream DNS Provider. To use your own, select Custom.
Google (ECS, DNSSEC)
OpenDNS (ECS, DNSSEC)
Level3
Comodo
Quad9 (filtered, DNSSEC)
Quad9 (unfiltered, no DNSSEC)
Quad9 (filtered, ECS, DNSSEC)
Cloudflare (DNSSEC)
Custom
< OK >
< Exit >


Blocklists
Pi-hole relies on third party lists in order to block ads.
You can use the suggestion below, and/or add your own after installation.
Select 'Yes' to include:
StevenBlack's Unified Hosts List
< Yes >
< No >


Enable Logging
Would you like to enable query logging?
< Yes >
< No >



Step 5: Verify Pi-hole

Use the following terminal command:

```
sudo pihole status
```

Step 6: Set admin password

Use the following terminal command:

```
sudo pihole setpassword 'Welkom#01'
```

Step 7: Configure Pi-hole and LAN clients

See assignments document chapter 4 paragraphs 4.3 and 4.4

Week 5 - Mac

No differences

Week 6 - Mac

No differences